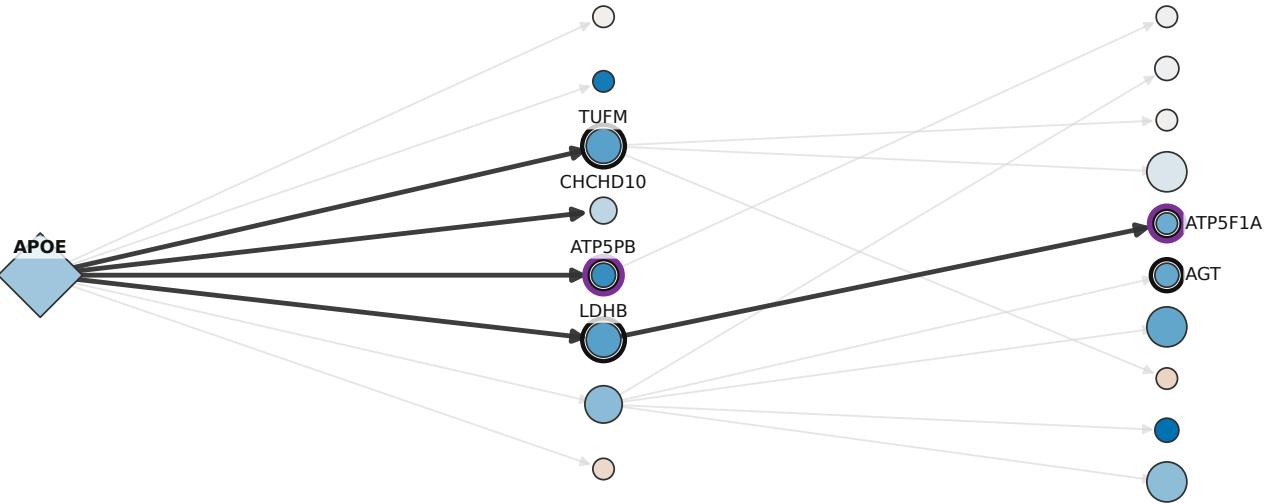
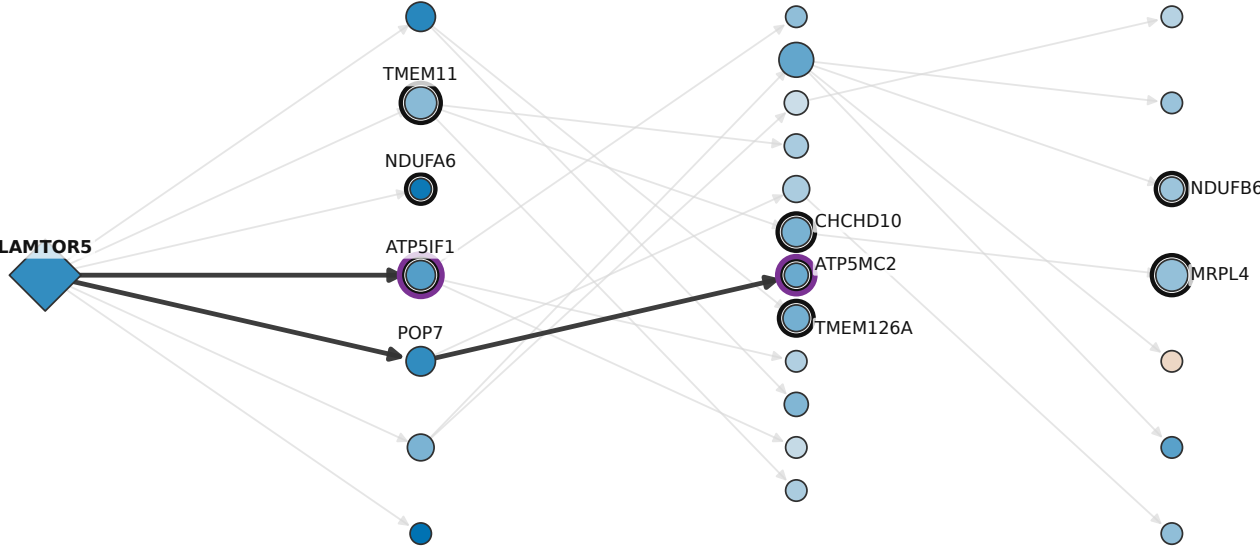


Directed KDA neighborhoods connect key drivers to mitochondrial and ATP-synthase genes

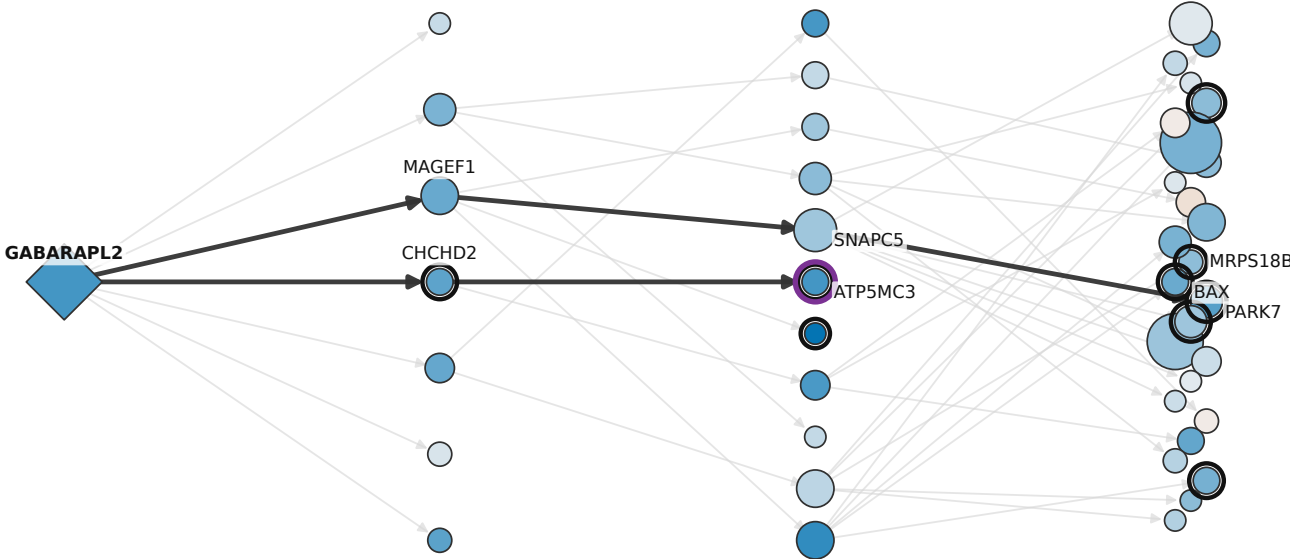
A Astrocyte APOE neighborhood
Ast GRM3 | Male | APOE e2 | AD-down mitochondrial signature
L=2 | n=19 | overlap=5 | FE=17.1 | q=0.00041



B Excitatory-neuron LAMTOR5 neighborhood
Exc L3-4 RORB CUX2 | Male | APOE e2 | AD-down mitochondrial signature
L=3 | n=27 | overlap=8 | FE=14.0 | q=1e-05



C Excitatory-neuron GABARAPL2 neighborhood
Exc L4-5 RORB GABRG1 | Male | APOE e2 | AD-down mitochondrial signature
L=3 | n=45 | overlap=9 | FE=9.9 | q=3.4e-05



Visual encodings

- ◇ Key driver
 - KDA overlap gene
 - ATP synthase / Complex V
 - Highlighted directed path
 - Other directed edge
- Node fill: AD vs NCI logFC
blue = lower in AD; orange = higher in AD
white = unavailable
- Node diameter: full-network total degree (linear)
- Black outer ring: gene is in the KDA overlap
Purple outer ring: MitoCarta Complex V gene

Together, the selected candidate systems connect mitochondrial translation and stress-control genes to ATP-synthase components.

